Building a User Simulator for Movie Recommendation: A Comparative Study of Static and Sequential Models

Course	CS 6140 Machine Learning - Final Project
Instructor	Prof. Smruthi Mukund
Team	Bolai Yin | Junke Zhu | Peihan Wang
Date	April 2026

Executive summary

This white paper studies whether the chronological order of a user's watch history adds predictive value beyond static preference features on MovieLens 1M. We compare an XGBoost simulator built on latent collaborative-filtering features with a GRU4Rec simulator that models ordered sequences. The updated sequential notebook used for the presentation treats GRU4Rec with SVD-initialized, unfrozen embeddings as the main sequential branch, while a frozen-SVD version is retained as a control. Across ablations, diagnostics, and segment-level analysis, XGBoost remains the strongest overall simulator on this dataset. The final tuned XGBoost model reaches test AUC 0.7997, while the presentation-aligned GRU4Rec-SVD unfrozen model reaches about 0.781 in the final comparison. The evidence supports a careful conclusion: under this dataset and evaluation setup, MovieLens 1M contains limited additional sequential signal beyond stable user-item preference structure.

1. Objectives and significance

A user simulator predicts the probability that a user will like a recommended item, written here as P(like | user, movie). In simulator-based recommendation, this probability model can stand in for real users during offline reinforcement learning, where direct online experimentation is slow, expensive, and risky.

Our project asks a focused machine learning question: does watch-order information provide meaningful predictive signal beyond aggregate preference information? To answer that question, we compare two deliberately different modeling paradigms. XGBoost treats the problem as structured binary classification over latent factors and metadata, while GRU4Rec treats the same problem as sequence modeling over ordered movie histories.

This version of the report also adopts several strengths from the Claude-generated draft: a clearer academic structure, a stronger background section, more explicit discussion of calibration as simulator quality, and fuller limitations. At the same time, all sequential-model conclusions and figure notes below have been refreshed to match the current notebook used in the presentation, cs6140_Final_Project_v2_update(2).ipynb.

2. Background

XGBoost is a gradient-boosted tree method that is particularly effective on tabular data with heterogeneous features. In this project it operates on low-dimensional user and item latent vectors together with movie metadata and user demographics.

GRU4Rec uses gated recurrent units to summarize a user's chronological interaction history into a hidden state. It is therefore a natural candidate when short-term intent, recency, or order effects may matter.

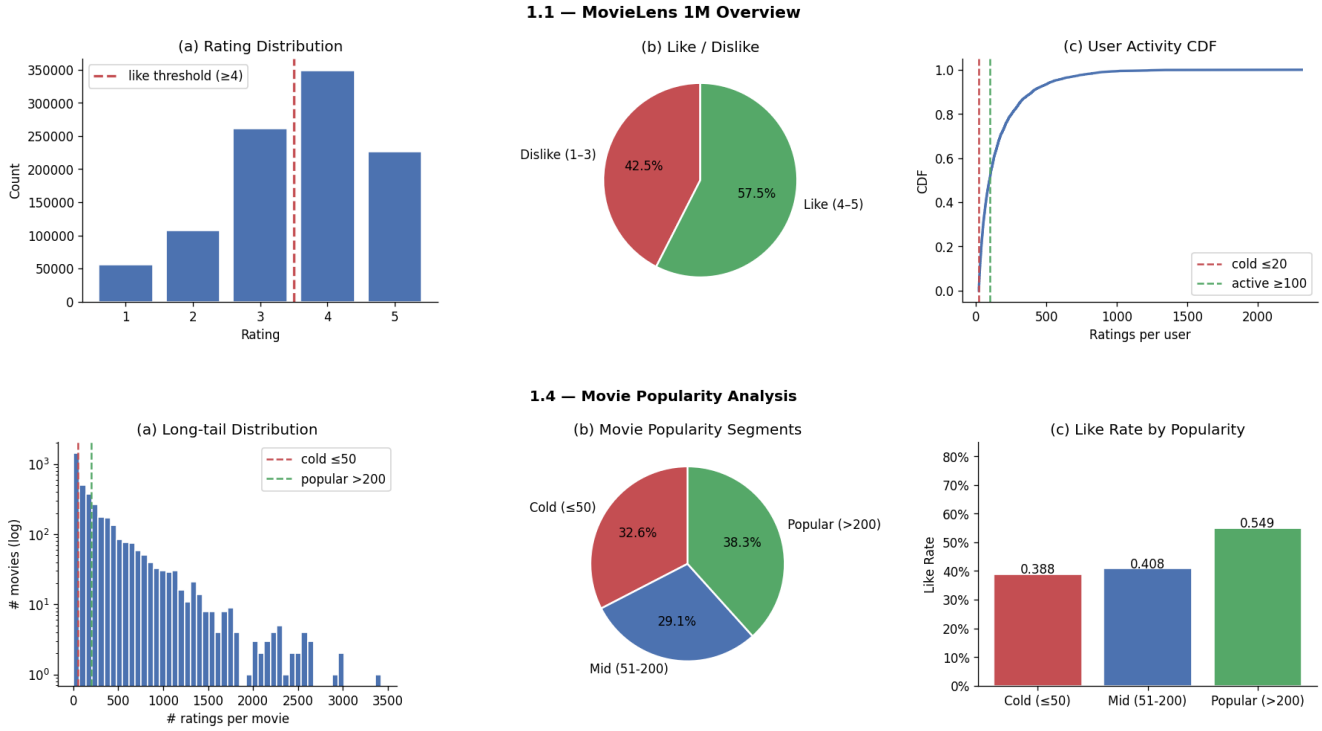
TruncatedSVD plays an important supporting role in both branches of the project. In the XGBoost branch, SVD user and item vectors provide dense collaborative-filtering features. In the GRU branch, SVD item embeddings help isolate whether stronger item representation alone can narrow the gap with the static model.

3. Data and preprocessing

Ratings	Users	Movies	Positive rate
1,000,209	6,040	3,706	57.5%

Ratings were binarized using rating >= 4 as the positive class. The implementation uses a per-user chronological split: for each user, the earliest 80 percent of interactions are assigned to training, the next 10 percent to validation, and the final 10 percent to test. This design keeps the evaluation temporally honest and avoids the optimistic leakage that would arise from a naive random split. Figure 1 summarizes the dataset profile and the main EDA views used to motivate the modeling setup.

All training-derived artifacts were computed from the training portion only. That includes the rating matrix used for SVD, movie popularity counts, and any aggregate statistics used for downstream features. This design helps preserve a clean causal direction from past behavior to future prediction.



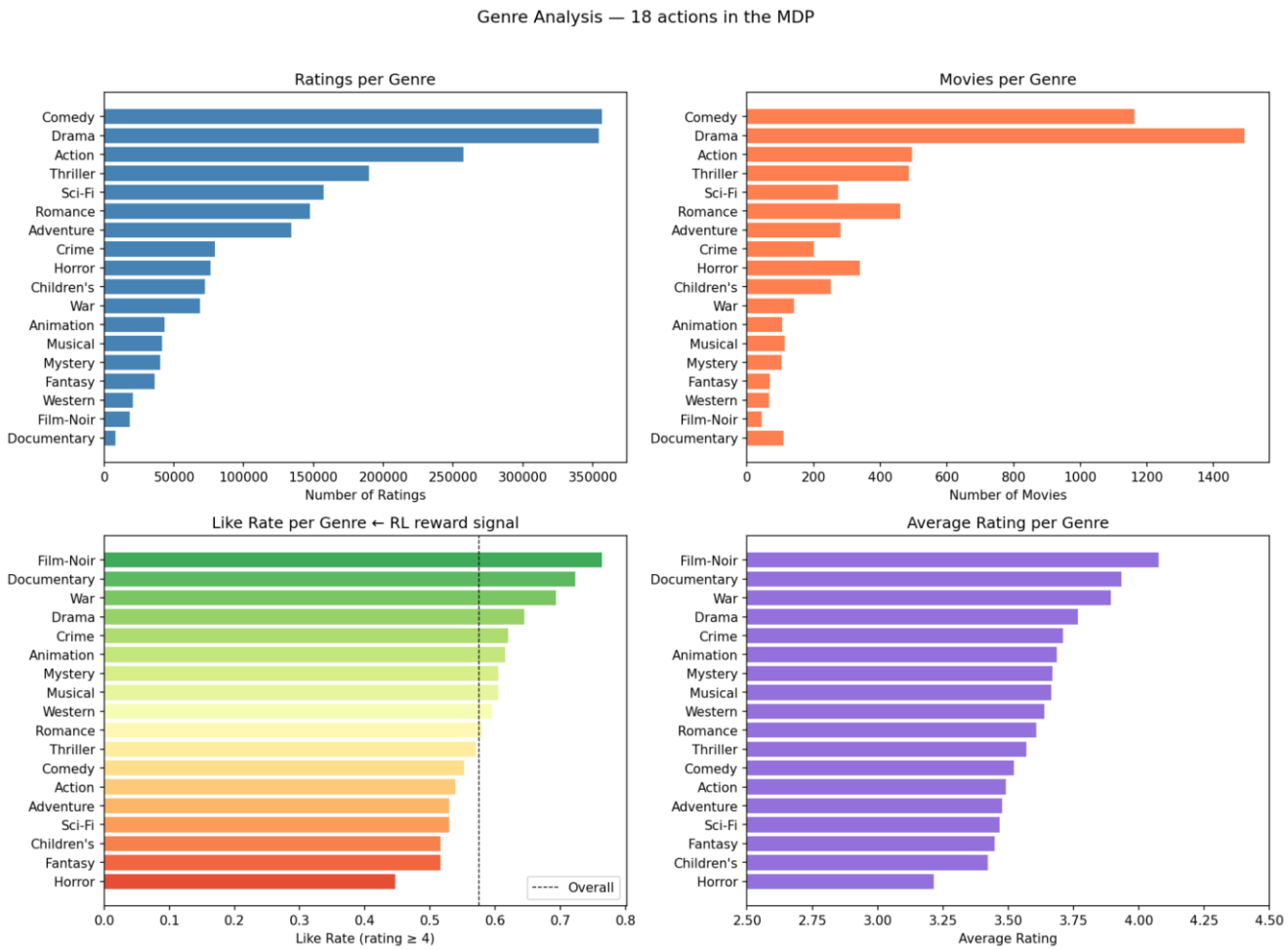


Figure 1. EDA overview and data profile

4. Methods

4.1 XGBoost simulator

The XGBoost branch models each user-movie pair with structured features. The final presentation version describes a feature set of 88 dimensions: 32-dimensional user SVD, 32-dimensional item SVD, 18 genre indicators, release year, number of genres, user demographics, and movie popularity derived from training data. In the ablation study, SVD features already provide most of the predictive signal, while metadata and demographics supply incremental gains.

The tuned XGBoost configuration uses max_depth = 6, learning_rate = 0.05, and n_estimators = 500. Additional sweeps examine SVD dimension, tree depth, regularization, and number of estimators. The report treats XGBoost as the main static baseline because it directly answers the research question: how much can be achieved when sequence order is ignored?

4.2 GRU4Rec simulator

The GRU4Rec branch consumes a chronological sequence of watched movie identifiers and predicts the probability that the user will like a candidate target movie. The baseline version uses 64-dimensional learnable item embeddings and a GRU hidden size that is varied in ablation. The main sequence-length experiments

sweep T from 5 to 100. Training uses binary cross-entropy with Adam, dropout, and early stopping on the validation set.

The updated notebook adds two SVD-initialized variants. In the frozen-SVD version, item vectors from the training-set SVD are fixed during training, isolating the effect of stronger collaborative initialization without adaptation. In the unfrozen-SVD version - which is the branch used in the final presentation - the same SVD initialization is allowed to fine-tune through backpropagation. This makes the ablation more informative: if unfrozen SVD clearly beats both learnable and frozen variants, then pretraining plus adaptation matters; if the gain over frozen is small, the remaining bottleneck is more plausibly weak sequential signal or information loss from GRU history compression.

4.3 Evaluation strategy

Metric	Why it matters	How it is used here
AUC	Threshold-free discrimination	Primary comparison metric across all models
Log loss	Quality of probability estimates	Used to compare probabilistic fit and penalty for confident mistakes
ECE	Calibration	Important because simulator probabilities may later be used as reward signals
Single-threshold F1	Snapshot at decision metric	Included for completeness and slide-level reporting

Beyond headline metrics, the notebooks also examine calibration curves, SHAP-based feature interpretation, cold-versus-active user segments, and time-series cross-validation stability. This richer diagnostic layer is one of the strongest aspects of the project because it moves the analysis beyond a single leaderboard number.

5. Results

5.1 Exploratory analysis

The exploratory analysis gives an early clue about why a static model might do well. User activity is highly uneven, with a small number of highly active users and a long tail of lighter users. Popular movies also receive substantially higher like rates than cold movies, showing that global exposure and popularity remain important background signals. Most importantly for the research question, the fatigue probe does not reveal a strong monotonic decline in ratings over repeated same-genre runs, which weakens the case that strong short-term sequential effects exist in this dataset (Fig. 2).

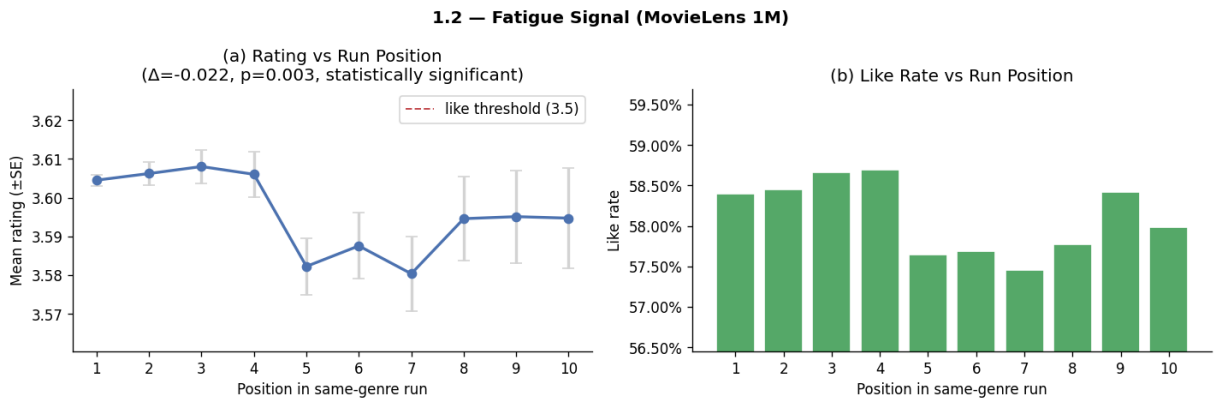


Figure 2. Fatigue signal and user activity findings

5.2 XGBoost ablations and diagnostics

Feature scheme	Dimensions	AUC
SVD only	64	0.7995
+ movie metadata	84	0.7997
+ user demographics	87	0.8003
+ popularity	88	0.7997

The ablation pattern is consistent with the project narrative. SVD features carry the bulk of predictive strength, movie metadata adds a small improvement, user demographics add the largest remaining marginal gain, and popularity contributes little once the personalized latent structure is already represented (Fig. 3). This is a useful result because it suggests the model is not merely exploiting popularity shortcuts.

In the final tuned pipeline, a full-feature XGBoost variant reaches test AUC 0.7997, log loss 0.5447, F1 0.7391, and expected calibration error about 0.030. Importantly, this result should be interpreted together with the ablation: closely related XGBoost variants perform at nearly the same level, so the main conclusion is not that popularity materially improves performance, but that latent user-item structure already explains most of the predictive signal. SHAP ranks user and item latent dimensions ahead of other features, while the cold-user gap remains visible: users with less interaction history are harder to model than active users (Fig. 4). Time-series cross-validation further shows that the result is stable across temporal folds rather than being driven by a single lucky split (Fig. 5).

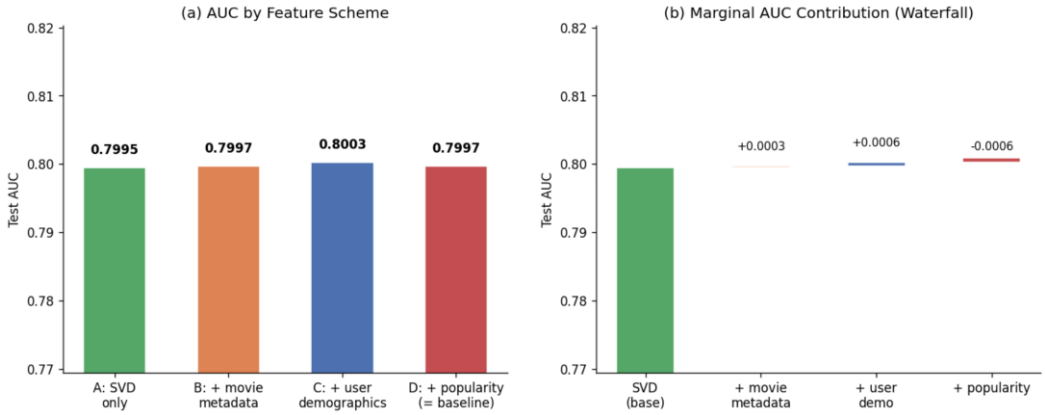


Figure 3. XGBoost feature-group ablation

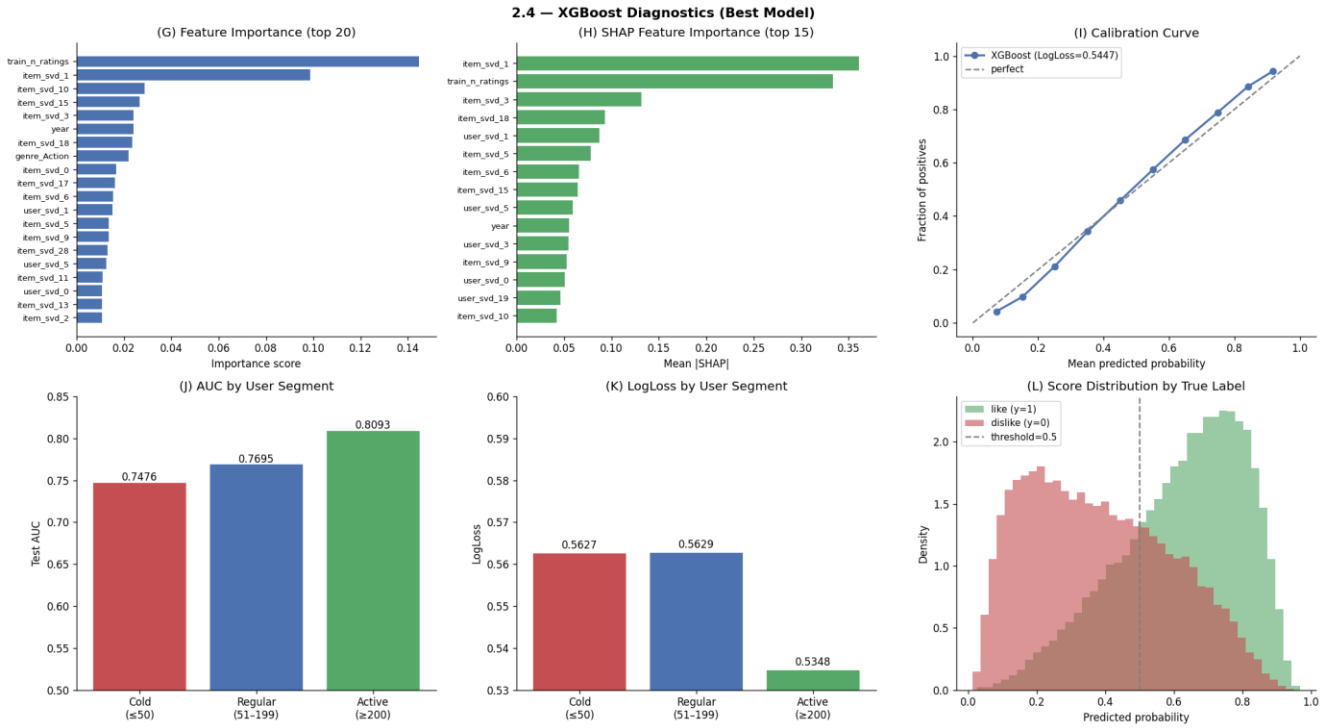


Figure 4. XGBoost diagnostics panel

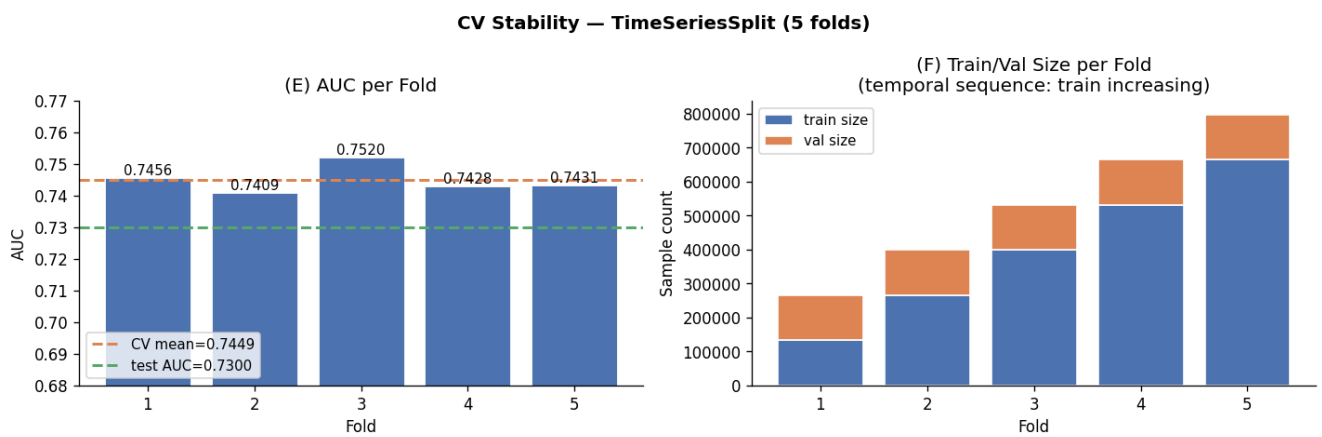


Figure 5. Time-series cross-validation stability

5.3 GRU4Rec ablations

Configuration	Hidden	T	AUC
Learnable embedding (random init)	128	100	0.7626
SVD frozen	128	100	0.7677
SVD unfrozen	64	100	0.7795
SVD unfrozen	128	100	0.7805
SVD unfrozen	256	100	0.7794
SVD unfrozen	512	100	0.7797

GRU4Rec benefits from longer histories, and the updated notebook shows that this benefit is much stronger for the SVD-initialized branch that is allowed to fine-tune. In the sequence-length sweep, frozen SVD improves

from test AUC 0.7398 at T = 5 to 0.7677 at T = 100, while unfrozen SVD rises from 0.7420 to 0.7830. The biggest gains appear at longer histories, especially between T = 20 and T = 50, which suggests that the sequential model does benefit from richer context - but still does not overtake the best static XGBoost model (Figs. 6 and 8).

The updated embedding ablation changes the interpretation in an important way. Frozen SVD is no longer the main sequential comparison target; instead, it serves as a control against the presentation-aligned unfrozen-SVD model. The final picture is: learnable random initialization is weakest, frozen SVD improves on it, and unfrozen SVD performs best among the GRU variants (Fig. 7). At the same time, the improvement from unfreezing over freezing is modest at short sequences and remains smaller than the residual gap to XGBoost, so the central project conclusion stays the same: stronger embeddings help, but they do not erase the dataset's preference for static latent matching.

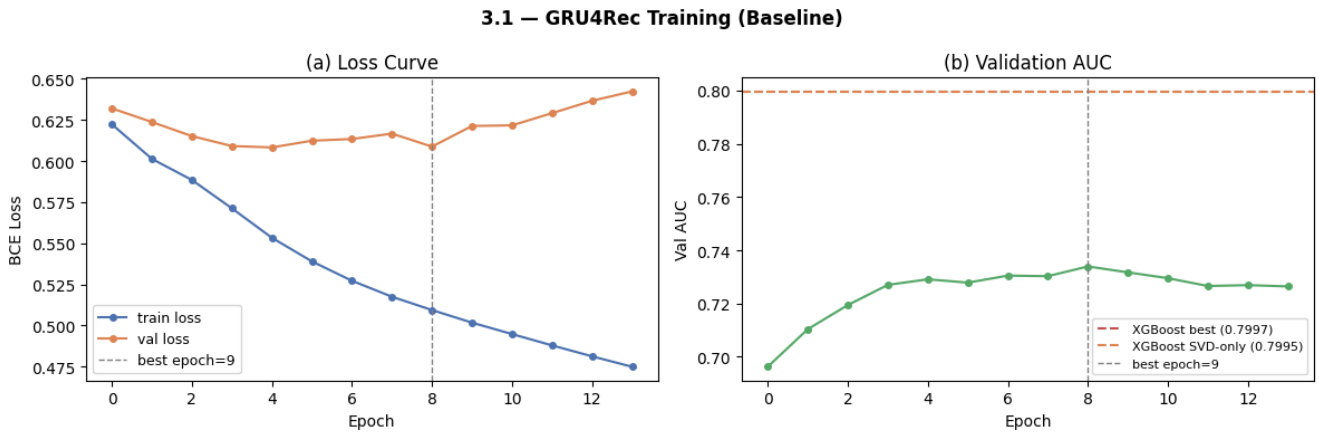


Figure 6. GRU4Rec baseline learning curves

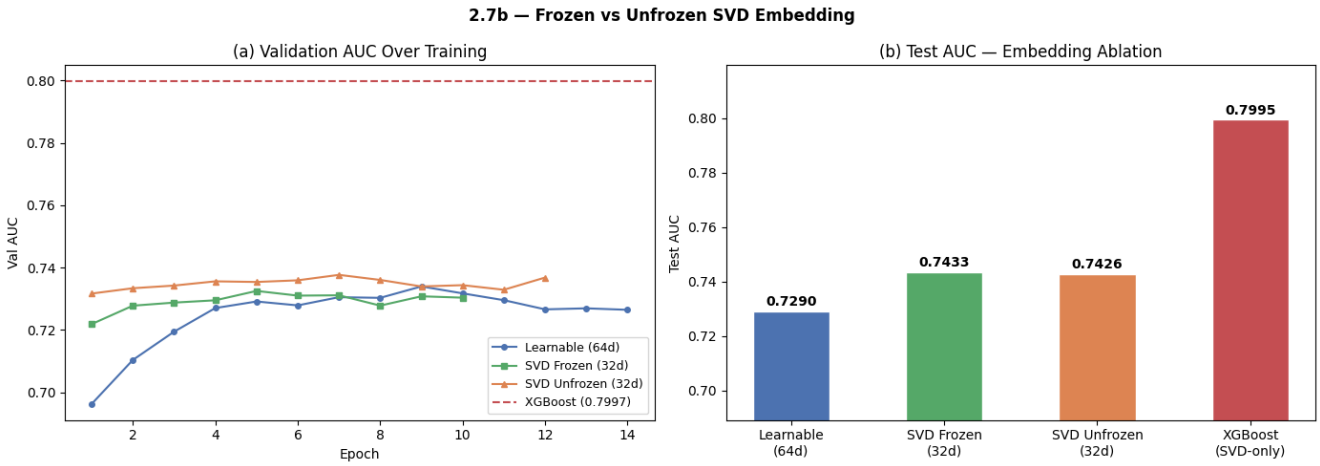


Figure 7. Embedding ablation: learnable vs frozen vs unfrozen SVD

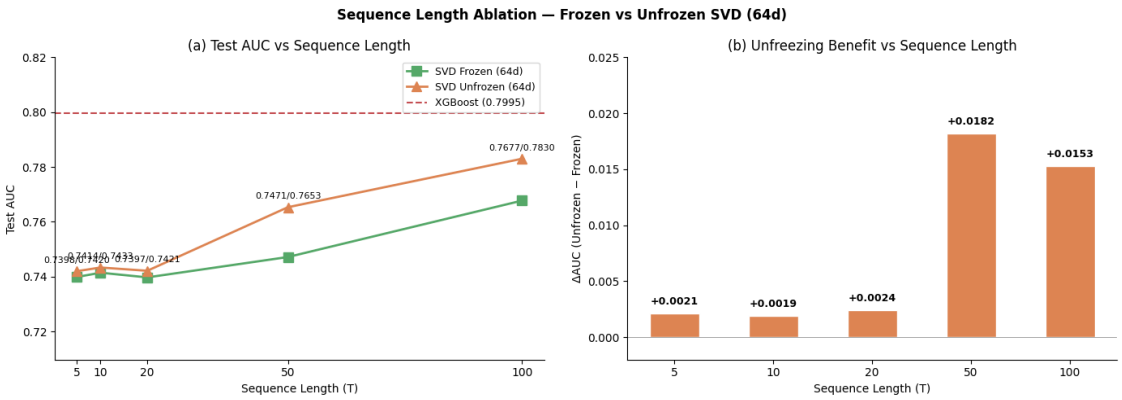


Figure 8. Sequence-length and model-capacity ablations for GRU4Rec

5.4 Final cross-model comparison

Model	Test AUC	Log loss	ECE	Train time
XGBoost (best tuned)	0.7997	0.5447	0.030	39 s
XGBoost (SVD-only reference)	0.7995	0.5449	0.030	35 s
GRU4Rec (learnable)	0.7577	0.5855	0.019	391 s
GRU4Rec-SVD (frozen)	0.7692	0.5734	0.0119	720 s
GRU4Rec-SVD (unfrozen)	0.7813	0.5713	0.0406	415 s

The final comparison answers the central project question. XGBoost remains the strongest overall simulator on MovieLens 1M by discrimination and training efficiency. In the updated Phase 5 notebook, the final tuned full-feature XGBoost variant reaches 0.7997 test AUC, while the directly matched SVD-heavy XGBoost reference reaches 0.7995. The small difference between these two static variants reinforces the ablation result: latent collaborative structure is doing most of the work, and extra popularity-style features add little marginal benefit. Against that static reference, GRU4Rec with learnable embeddings reaches 0.7577 test AUC, GRU4Rec-SVD frozen reaches 0.7692, and GRU4Rec-SVD unfrozen reaches 0.7813. So the presentation-aligned unfrozen-SVD model is the best sequential branch, but it still trails the static model by roughly 0.018 AUC (Fig. 9).

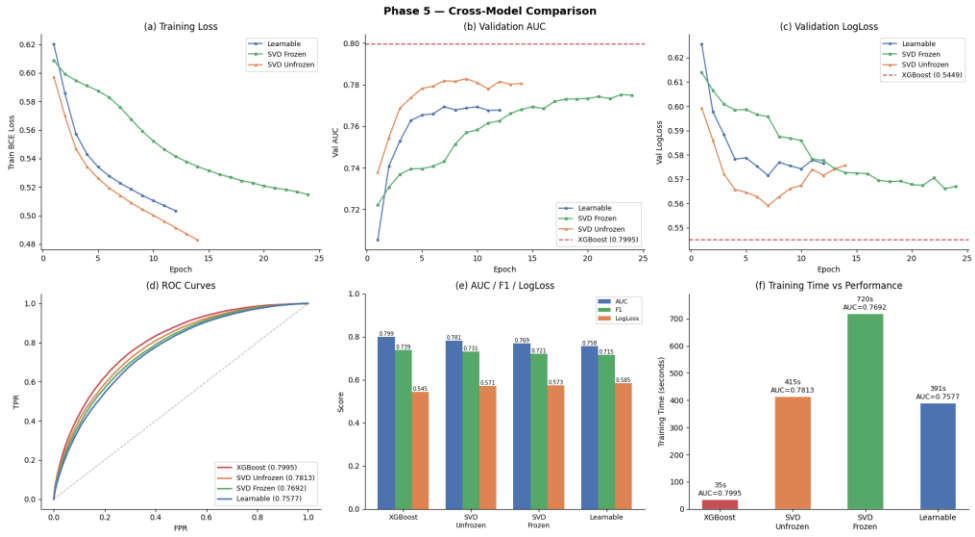


Figure 9. Cross-model comparison summary

5.5 Extended diagnostics and segment analysis

Several secondary analyses make the interpretation more robust. Per-user segment results show that all methods struggle on cold users and improve with richer histories. The updated notebook also sharpens the calibration story: GRU4Rec-SVD frozen is the best-calibrated sequential model with ECE 0.0119, GRU4Rec learnable is next at 0.0190, XGBoost is 0.0304, and GRU4Rec-SVD unfrozen is worst-calibrated at 0.0406 despite having the highest GRU AUC. This matters for simulator design: the best classifier is not automatically the best probability model. Score-distribution plots further show that unfrozen SVD is the most decisive GRU variant, with the largest separation between like and dislike predictions, while the frozen model offers a better reliability-accuracy trade-off (Fig. 10).

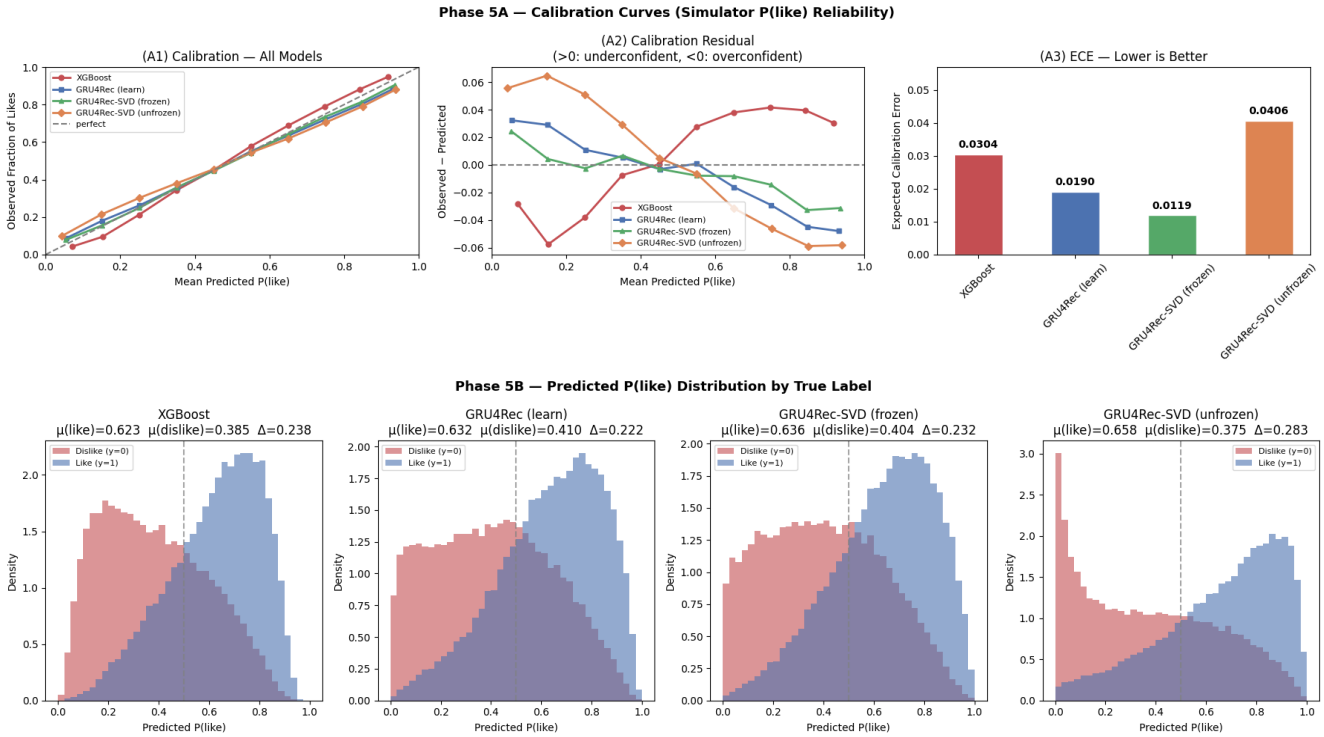


Figure 10. Cross-model calibration, score distributions, and per-user segment analysis

5.6 Cold-start setup illustration and evaluation protocol

To test robustness beyond the standard temporal split, we also ran strict cold-start evaluations in which held-out users or movies were removed entirely from training. This creates a harder and more realistic setting than ordinary future-interaction prediction, because the model must generalize with limited or no collaborative support for the held-out entity. Figure 11 summarizes the seen-versus-cold comparison for both XGBoost and GRU-based simulators.

The main result is that both model families degrade under cold-start, but the drop is much larger for unseen movies than for cold users. In the presentation-aligned comparison, XGBoost declines from roughly 0.80 AUC on seen data to about 0.74 for held-out users and about 0.72 for held-out movies, while GRU4Rec drops from about 0.76 on seen data to about 0.62 on held-out movies. Even after degradation, XGBoost remains clearly above random, whereas the sequential branch is much more brittle on truly unseen items (Fig. 11).

This pattern is consistent with the earlier ablations. XGBoost can still fall back on static metadata and user-level structure when collaborative evidence becomes sparse, whereas GRU4Rec depends more strongly on

learned item representations and observed interaction histories. Under strict unseen-movie holdout, that dependence becomes a liability: the sequential model has less opportunity to benefit from the very ordering signal it was designed to exploit. In that sense, the cold-start experiment strengthens the broader conclusion of the project. On MovieLens 1M, stable preference structure is more valuable than sequence modeling not only in the standard split, but also under harsher generalization conditions (Figs. 9 and 11).

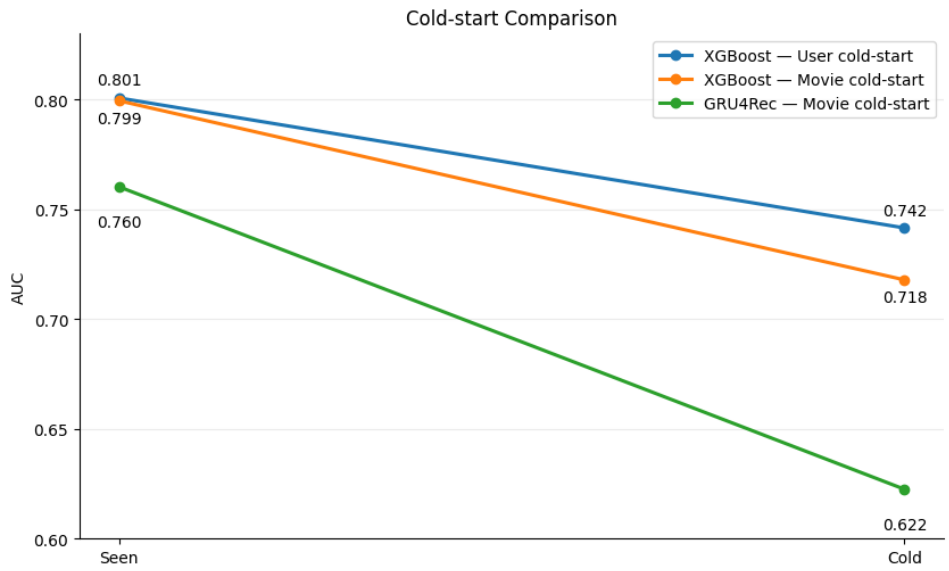


Fig 11. Strict cold-start comparison for XGBoost and GRU4Rec - Strict holdout cold-start comparison. Held-out users or movies are removed entirely from training. Both models degrade, but unseen-movie generalization is markedly harder, especially for GRU4Rec. XGBoost remains more robust under both cold-user and cold-movie settings.

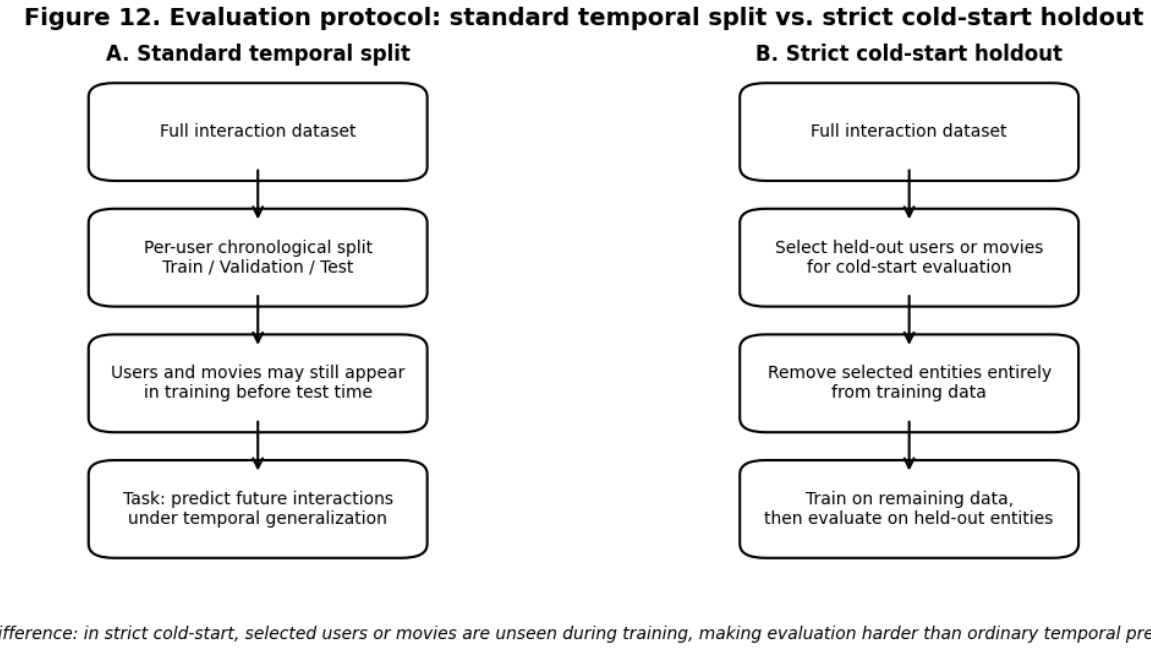


Figure 12. Cold-start setup illustration and evaluation protocol

Figure 12 illustrates the difference between the standard per-user temporal split and the stricter cold-start protocol. In the latter, selected users or movies are removed entirely from training and appear only at evaluation time, making the task substantially harder than ordinary temporal generalization.

6. Discussion

The most defensible interpretation is not that sequential recommendation models are inherently inferior, but that this particular dataset rewards stable preference matching more than short-term sequence reasoning. MovieLens 1M is a user-initiated ratings dataset rather than a high-frequency feed or session dataset. Under that regime, collaborative latent structure already explains much of the future preference signal. The updated notebook nuances that story: sequential modeling is not entirely flat - unfreezing SVD and using longer histories pushes GRU4Rec substantially above the plain learnable baseline - but the gains remain insufficient to beat the best static approach (see Figs. 7-10).

This report also keeps one of the Claude draft's best ideas: treating calibration as part of simulator quality rather than an afterthought. The updated results make that point even more concrete. The highest-AUC GRU variant is the unfrozen-SVD model, but the best-calibrated GRU variant is the frozen-SVD control. If the team extends the simulator into reinforcement learning, that distinction could matter for reward stability, off-policy evaluation, and safe policy comparison. In other words, better discrimination and better calibration do not necessarily come from the same model choice, as shown most clearly in Fig. 10.

7. Limitations and future work

- Dataset limitation: MovieLens 1M is not optimized for session dynamics, repeated exposure, or platform-driven fatigue. It is therefore a relatively weak test bed for sequence-sensitive recommendation.
- Evaluation limitation: the sequential branch uses train-available history construction for validation and test. This keeps the split clean but may understate the benefit of continually updated online context.
- Model limitation: the sequential comparison focuses on GRU4Rec.
- Cold-start limitation: both collaborative latent factors and sequence models degrade when user or item history is sparse, and strict unseen-movie holdout is particularly challenging. In our current setup, unseen movies are represented mainly through limited metadata, which constrains generalization beyond collaborative support.

Future work should test the same comparison on denser temporal datasets such as KuaiRec or Tenrec, add an attention-based sequential model, and evaluate whether the best-calibrated simulator leads to better downstream reinforcement-learning behavior than the highest-AUC simulator. The current figures already suggest this trade-off is real: the most accurate sequential model and the most reliable sequential probability model are not the same system (Figs. 9-10).

8. Team contributions

Bolai Yin led project architecture, XGBoost modeling, ablation design, diagnostics, and cross-model comparison framing.

Junke Zhu implemented the GRU4Rec models and the sequential training pipeline, including the learnable-embedding, frozen-SVD, and unfrozen-SVD variants.

Peihan Wang led data preprocessing, feature engineering support, experiment tracking, and exploratory analysis.

The project was developed collaboratively through shared notebooks, iterative ablation design, and joint interpretation of results.

References

1. Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.
2. Harper, F. M., & Konstan, J. A. (2015). The MovieLens datasets: History and context. ACM Transactions on Interactive Intelligent Systems, 5(4), 1-19.
3. Hidasi, B., Karatzoglou, A., Baltrunas, L., & Tikk, D. (2016). Session-based recommendations with recurrent neural networks. International Conference on Learning Representations.
4. Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. Proceedings of the 34th International Conference on Machine Learning.
5. Koren, Y., Bell, R., & Volinsky, C. (2009). Matrix factorization techniques for recommender systems. Computer, 42(8), 30-37.
6. Team project notebooks, slides, and experimental logs for CS 6140 Machine Learning, Spring 2026.